

Radiology Report Generation with Vision Transformers and Retrieval-Augmentation

Madhumitha V, Sriharshini Dusi, Yueqiao Chen

1 Introduction

Automated radiology report generation is a promising application of multimodal deep learning, aiming to produce clinically accurate textual descriptions from medical images such as chest X-rays. By combining computer vision and natural language processing (Vision + NLP), these systems can help reduce diagnostic workload, improve consistency, and support clinical decision-making. In this project, we propose a multimodal architecture that uses Vision Transformers (ViT) as the image encoder and a Transformer-based decoder to generate natural language reports. ViTs are compared with traditional CNN-based encoders to evaluate their effectiveness in capturing visual semantics and improving report quality. To further enhance semantic grounding, we incorporate retrieval-augmented learning, where top-k semantically similar image-report pairs from the training set are retrieved using FAISS and used as contextual input during generation. Our experiments are conducted on the IU X-Ray dataset, which contains 7,470 chest X-rays paired with diagnostic reports. We evaluate model performance using standard language generation metrics (BLEU, ROUGE, CIDEr), and assess semantic alignment with the input image using CLIPScore.

1.1 Research Questions

1. **How does a Vision Transformer (ViT)-based multimodal encoder perform compared to a CNN-based image encoder for radiology report generation? Research Question 1** investigates whether Vision Transformer (ViT) based encoders can outperform traditional convolutional neural network (CNN) based encoders in generating clinically coherent and semantically rich radiology reports. Specifically, we aim to implement a ViT-based multimodal encoder that jointly processes both image and textual features and compare against a CNN-based image encoder. The evaluation will be conducted using standard NLP metrics to assess report quality.
2. **How does retrieval augmentation help with the semantic alignment of the image with the generated report? Research Question 2** focuses on the impact of retrieval augmentation in enhancing report generation quality, particularly in terms of improving image-text semantic alignment and reducing hallucinated content. We propose retrieving similar image-report pairs from the training dataset to condition both the encoder and decoder process.
3. **How can vision-language metrics like CLIP-Score be used to assess the semantic alignment of automatically generated reports? Research Question 3** explores whether vision-language similarity metrics can offer deeper insights into the quality of generated reports. To evaluate the outputs beyond standard NLP metrics, we investigate how semantic metrics like CLIP-Score can be employed to measure the semantic alignment between generated reports and their corresponding radiology images, providing a multimodal evaluation framework that captures clinical relevance more effectively.

2 Literature Background

Radiology report generation has become a rich field of research, blending innovations in computer vision and natural language processing. Among the pioneering contributions is the X-RGen framework by Chen et al. (2024) [1], which introduces a radiologist inspired architecture that mirrors the clinical reporting

workflow. X-RGen breaks down the report generation task into four stages: initial observation, cross-region analysis, medical interpretation, and report formation. This method serves as a strong baseline and does not incorporate Vision Transformers (ViTs) or retrieval mechanisms. Each phase is carefully designed to simulate how a radiologist interprets medical imagery. The image encoder is based on a ResNet-101 CNN that extracts feature maps, and the model further employs cross-region contrastive learning to enhance anatomical discriminability. Moreover, X-RGen integrates domain specific medical knowledge into the generation pipeline using topic modeling and a co-attention mechanism, improving the clinical relevance of the output.

Despite its strength in aligning with radiological practice and yielding competitive performance on benchmarks, X-RGen has two core limitations relevant to our work. First, it relies on CNNs, which are limited in capturing global visual context compared to ViTs. Second, it does not utilize retrieval mechanisms for knowledge grounding, potentially making it more susceptible to hallucinations or factual errors. Our project directly builds upon this baseline, investigating whether Vision Transformers can replace the CNN backbone to better model spatial dependencies and long-range features, and exploring retrieval augmented generation (RAG) to supplement the decoder with semantically relevant prior examples. This positions our work as a natural extension of X-RGen, aiming to preserve its radiologist inspired interpretability while enhancing visual encoding and factual grounding.

Complementing this, Li et al. (2025) provide a comprehensive survey of deep learning approaches in radiology report generation [3]. Their work categorizes the field into CNN-based methods, transformer-based models, contrastive learning approaches, and knowledge-augmented frameworks. This survey is particularly valuable for mapping the research landscape and identifying gaps in methodology, evaluation, and clinical applicability. The authors emphasize the need for better alignment between generated reports and real-world clinical reasoning, highlighting the importance of both semantic fidelity and explainability. They also call attention to challenges in evaluation, such as the inability of common metrics like BLEU and ROUGE to fully capture clinical correctness. Hence, we aim to explore additional evaluation metrics that better assess semantic correctness and clinical relevance in generated reports.

Further advancing this trajectory are recent multi-modal transformer models (2024) that pair ViT-based encoders like BEiT or DEiT with autoregressive decoders such as GPT-2 or DistilGPT for report generation [4]. These systems introduce cross-attention mechanisms that allow language models to attend directly to image features, enabling fluent, context-aware report synthesis. The use of large-scale pretraining on vision and text modalities helps these models generalize better and generate more coherent reports. In empirical benchmarks, these architectures often outperform traditional CNN+RNN or CNN+Transformer pipelines, particularly on metrics like ROUGE-L and CIDEr.

Retrieval-augmented generation (RAG) has been explored to enhance the factual correctness of radiology report outputs[5]. For example, recent studies[16] have demonstrated that integrating knowledge retrieval with large language models enables error detection, localization, and correction in clinical documentation, with retrieval mechanisms drawing on both internal and external sources to ground the model’s outputs in relevant medical knowledge. These approaches focus more on refining existing outputs than end-to-end generation. However, most transformer-based models in this line of work do not incorporate retrieval-based grounding. As a result, they can still produce semantically fluent but clinically incorrect content, especially when exposed to out-of-distribution samples or ambiguous findings. Our project seeks to extend this approach by combining ViT-based encoders with retrieval-augmented generation. We hypothesize that introducing retrieval mechanisms, bringing in similar image-report pairs from the training set will provide additional context and reduce hallucinations, especially in ambiguous cases.

Lastly, the R2Gen model (2021) combines a CNN encoder with a cross-modal memory network, aligning visual and textual modalities through memory-augmented mechanisms [6]. Though it lacks transformer-based encoders and retrieval components, R2Gen remains a strong benchmark for memory-based image-to-text generation.

Collectively, these works provide a comprehensive foundation for understanding the strengths and limitations of current radiology report generation methods and highlight the opportunities for our project,

which aims to integrate ViT and retrieval augmentation for improved semantic alignment and factual precision.

3 Data and Methodology

3.1 Dataset

We base our study on the task of automated radiology report generation, which aims to produce coherent and clinically accurate diagnostic reports from chest X-ray images. The IU X-Ray dataset[2] is a widely adopted benchmark in this domain, containing both frontal and lateral images. It contains 7,470 chest X-ray images paired with 3,955 de-identified free-text reports. Each report includes structured sections such as Findings and Impression. The dataset is divided into training (2069 samples), validation (296 samples), and test (590 samples) sets after filtering out cases with only a single image in accordance with preprocessing techniques. The IU X-Ray dataset also includes metadata tags, as shown in Figure 2, that define clinical conditions and anatomical features observed in the radiographs, such as "degenerative change," "cardiomegaly," or "pleural effusion." These tags can be useful for auxiliary tasks like image classification or retrieval because they offer weak supervision signals. The uneven tag distribution reflects the clinical prevalence. In the real world, certain conditions show up significantly more frequently than others.

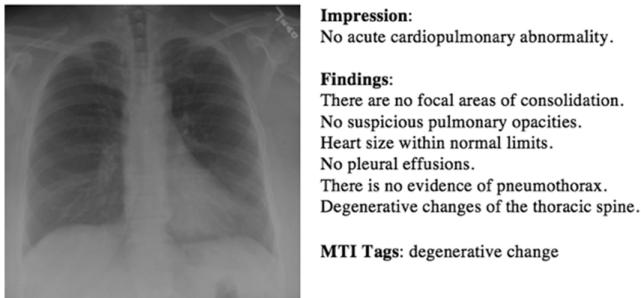


Figure 1: *Example Radiology Report*

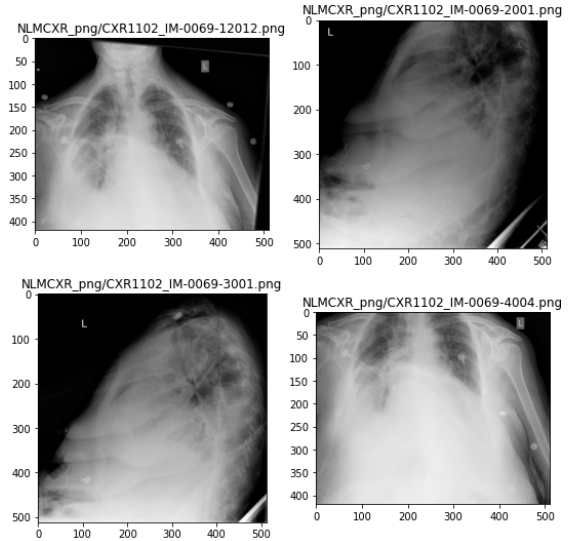


Figure 2: *Tag Distribution in IU X-Ray Dataset*

3.2 Approach

Our approach builds upon the X-RGen [1] baseline with the goal of enhancing visual-textual alignment in radiology report generation. We explore two key extensions: (1) integrating a Vision Transformer (ViT) as the visual encoder and (2) incorporating a retrieval-augmented generation (RAG) mechanism to enrich report generation. These design choices are motivated by specific limitations observed in existing systems.

First, while the original X-RGen model offers strong domain-specific architectural priors through radiologist-inspired generations stages, its use of ResNet limits the capacity to capture long-range dependencies or holistic image-level patterns. We hypothesize that ViT, with its global self-attention mechanism, can provide more semantically rich and globally inherent image features, potentially leading to better grounding of generated reports. However, we also recognize that ViT alone may not replicate the structural guidance embedded in X-RGen. Thus, instead of treating ViT as a full replacement, we explore

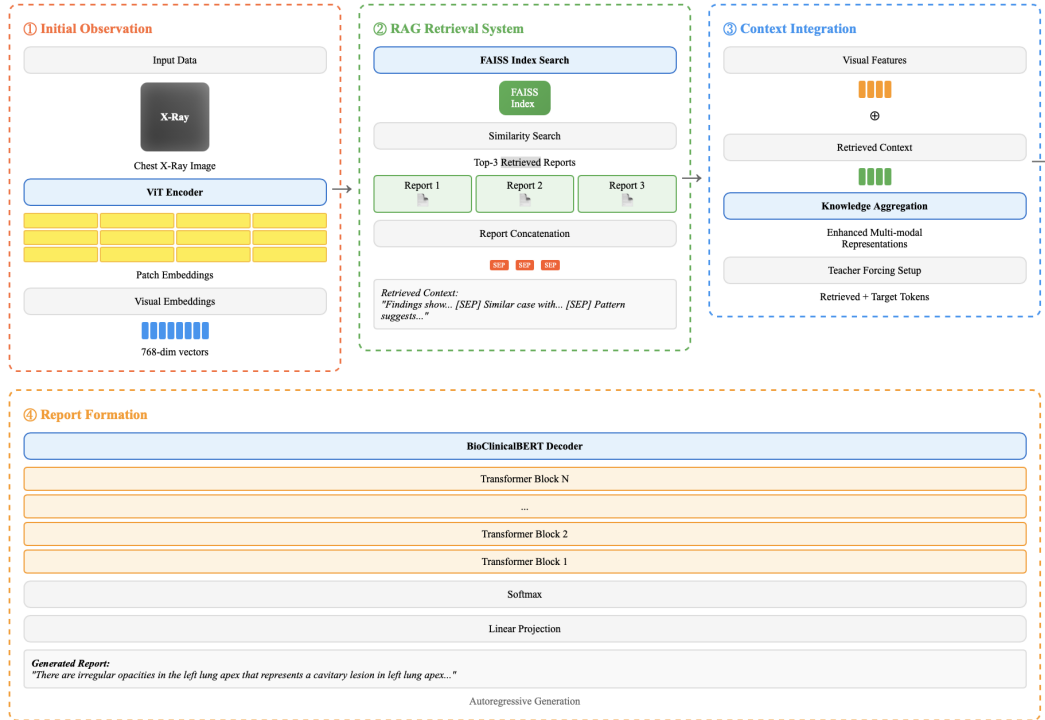


Figure 3: *RAG Integration Pipeline*

integrating it within the X-RGen framework, retaining its modular decoder stages while upgrading its visual backbone.

Second, motivated by recent progress in retrieval-augmented generation, we explore whether incorporating semantically similar image-report pairs, as shown in Figure 3, could help guide the decoder during training. Radiology reports often contain subtle, case-specific variations, and conditioning the model with relevant retrieved examples could make generation more accurate and context-aware. We chose RAG as a lightweight and modular way to test this hypothesis. That said, we were also aware of potential pitfalls, such as retrieval noise and input length constraints, and treated this as a complementary direction to ViT-based improvements.

Third, we also investigate whether visual data augmentation can further improve performance when applied to the best-performing model. We apply commonly used augmentation techniques such as random resized cropping (scale range: 0.8-1.0), resizing to a fixed resolution (224x224), and color jittering with saturation variation of a range of 20%. These transformations aim to introduce visual variability while preserving diagnostic content. Although such augmentations are standard in natural image tasks, their utility in medical imaging, where fine-grained pixel-level cues are critical, is less established. Our goal is to assess whether controlled augmentation enhances model generalization or instead disrupts the subtle visual-textual alignment necessary for accurate report generation.

Finally, we investigate MedViT, a ViT variant pretrained on biomedical images, to understand whether domain-specific pretraining offers advantages over generic ViT. Given the unique visual language of radiographs, we hypothesize that MedViT could provide a stronger initialization, particularly when integrated into structured generation pipelines like X-RGen.

Together, these explorations aim to bridge two dimensions that often seem in isolation: architectural structure (as in X-RGen) and model-scale representational power (as in ViT and RAG). Our goal is not merely to test performance gains, but to understand how these design components interact in the context of complex medical text generation.

3.2.1 Baseline

We chose Chen et al.[1] as the baseline in which the authors use X-RGen architecture, as depicted in Figure 5, which breaks down the report generation process into four radiologist-inspired stages: initial observation, cross-region analysis (CA), medical interpretation (MI), and report formation. The image encoder simulates a radiologist’s initial scan by extracting convolutional feature maps using ResNet-101. By aligning image-report pairs across several anatomical regions during training, the model uses cross-region contrastive learning to improve recognition. In the medical interpretation phase, domain knowledge is introduced through a co-attention mechanism that fuses the image features with region-specific medical terminology curated via topic modeling and manual filtering. Lastly, these medically aware embeddings are used to generate reports via a Transformer-based decoder.

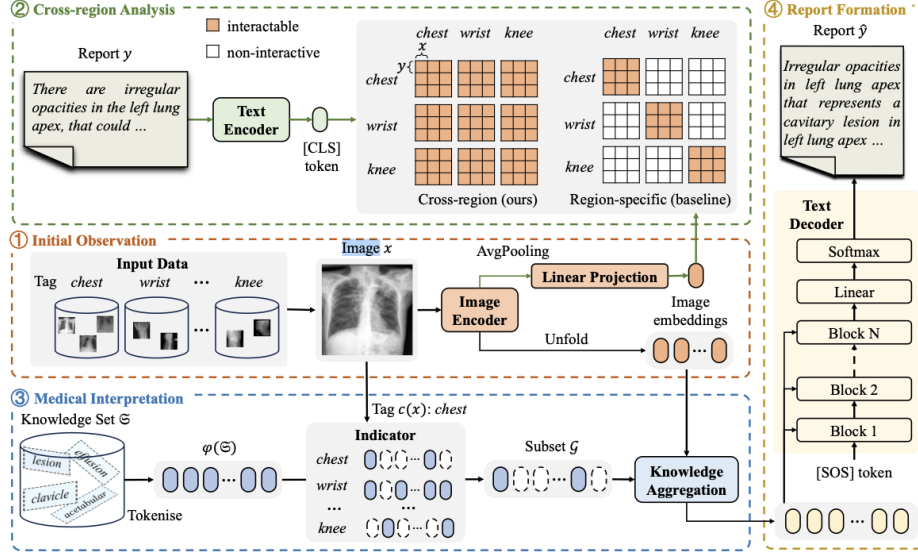


Figure 4: *Baseline Architecture*

We have successfully re-implemented the baseline model and obtained initial results, which will be discussed in detail in Section 5. This baseline serves as a reference point for evaluating future improvements. Our implementation focuses on generating radiology reports for chest X-ray images using the IU-Xray dataset, which contains only chest X-ray cases. This ensures the model is specialized for this modality. During training, we set a maximum of 100 epochs and applied early stopping to prevent overfitting. We used a batch size of 16 to accommodate GPU memory constraints. For evaluation, we implemented BLEU-1, BLEU-2, BLEU-3, BLEU-4, CIDEr, and ROUGE-L to enable comprehensive comparison and analysis of the generated reports. We also evaluate the semantic alignment of images and generation text using CLIP-Score.

3.2.2 Vision Transformer

We aim to replace the ResNet backbone in our current approach with a Vision Transformer (ViT), and MedViT, which are pretrained on medical or natural image datasets. Compared to CNNs, ViTs can better capture the overall structure and long-range relationships in images, an important feature for identifying complex or subtle conditions like effusion or cardiomegaly. We believe this change could improve how well the generated reports match the medical images.

To test this idea, we first built a simple model using a basic ViT encoder (ViT-Base) and a Transformer decoder (GPT2), without adding any retrieval components or dual-branch design. In this setup, we use the CLS token from the ViT as a summary of the image, then pass it through a linear layer to match the GPT2 input size. This processed image feature is used as context for the decoder during report generation.

We trained the decoder end-to-end on the IU X-Ray dataset using teacher forcing with paired image-report data, where the ground truth token at each step is fed into the model instead of its own prediction. Although this basic model does not use medical knowledge or retrieved examples, it gives us a useful baseline to understand how well a pure Transformer-based setup works.

However, we observed that the performance of the vanilla ViT encoder combined with a GPT-2 decoder was particularly lower than that of the baseline X-RGen model. To address this performance gap, we tried different approaches to improve the ViT frameworks. We suspected that the performance discrepancy is primarily due to differences in the architectural choices of the baseline model:

1. The baseline model initialized both its encoder and decoder with pre-trained weights specific to the medical domain, which likely provided a strong prior for the task.
2. The baseline employed BioClinicalBERT as the decoder, which is pre-trained on clinical notes and thus better suited for radiology report generation.

To address the limitations observed in the vanilla ViT+GPT2 architecture, we replaced the original CNN-based encoder with a MedViT model, which is pre-trained on medical imaging data. This substitution aimed to explore the impact of transformer-based visual representations on radiology report generation.

MedViT could better capture radiological patterns than its ImageNet-trained counterparts. Our architecture integrated the baseline X-RGen in terms of encoder-decoder design. When using MedViT, the model employs the large architecture (1024 embedding dimension, 24 transformer layers, 16 attention heads) rather than the base architecture. A projection head is used to align the patch embeddings and visual tokens from the CLS token with the textual embedding space, projecting from 1024 dimensions to 512 dimensions.

The decoder, which is a BioClinicalBERT-based Transformer decoder, is unaltered from the baseline. This enables the model to generate reports by leveraging biomedical language priors. The decoder generates free-text reports in an autoregressive manner and is dependent on the visual embeddings.

A checkpoint trained on medical image datasets is used to load pre-trained MedViT weights. The MedViT encoder produces a sequence of image patch embeddings. These embeddings are passed to a BioClinicalBERT-based Transformer decoder for cross-attention, which attends over visual tokens while generating text. As seen in Table 1, MedViT significantly improves CIDEr and ROUGE-L scores over vanilla ViT, indicating enhanced semantic alignment with ground-truth reports. We also observed that MedViT has better performance with CLIP-Score, as shown in Table 3.

3.2.3 Retrieval-Augmented Generation

We implemented a Retrieval-Augmented Generation (RAG) strategy into the baseline pipeline to enrich radiology report generation by incorporating semantically relevant examples from the same training set. The goal was to provide the model with additional domain-specific context at training time, thereby improving its ability to generate more informative and factually grounded results.

The RAG pipeline was built on top of a FAISS-based retrieval system. First, we extracted visual embeddings for all training images using the ViT encoder. Specifically, the image embeds vector from a projection head applied to the encoder’s output was used as the index key. These embeddings were stored in a FAISS flat index, and each entry was mapped to its corresponding radiology report. During training, for each input image, we extracted its visual embedding and performed a similarity search over the FAISS index to retrieve the top-3 nearest neighbors. The reports associated with these retrieved images were then concatenated using SEP tokens to delineate boundaries between different retrieved sources.

To integrate the retrieved reports into the generated pipeline, we tokenized the concatenated text and prepended the resulting token sequence to the ground-truth report tokens before feeding them into the Transformer-based decoder. This modified decoder input provided the model with both the visual

context and a textual memory of semantically related reports. To ensure sequence length constraints were respected, the ground-truth report was truncated if the combined length of retrieved and target tokens exceeded the maximum sequence length. The decoder was trained in a standard teacher-forcing setup using cross-entropy loss on the target report, while the encoder-decoder architecture was supervised with a contrastive loss when enabled. The overall model is trained end-to-end for report generation, with optional contrastive loss between image and text embeddings to further align visual-linguistic representations.

3.2.4 Evaluation

To comprehensively evaluate the quality of generated radiology reports, we adopt a suite of widely used natural language generation metrics that capture different aspects of linguistic quality.

1. **BLEU-1 to BLEU-4**

BLEU (Bilingual Evaluation Understudy) scores are widely used to assess the fluency and precision of generated text by measuring n-gram overlaps with reference texts. For example, BLEU-1 captures unigram precision, while BLEU-4 extends this to 4-gram sequences, enabling evaluation of phrase-level coherence. These metrics are particularly effective for detecting surface-level accuracy, such as whether commonly used medical terms or short phrases appear in the output. However, they are limited in capturing semantic similarity or paraphrased content.

2. **CIDEr (Consensus-based Image Description Evaluation)**

CIDEr evaluates the similarity between a generated report and a set of ground-truth reports by considering term frequency-inverse document frequency (TF-IDF) weighting of n-grams. It rewards consensus across multiple human references, placing greater emphasis on clinically relevant and less frequent words that are often critical in medical descriptions. This makes CIDEr especially appropriate for radiology report generation, where accurately reproducing uncommon but diagnostically important terms contributes more to clinical validity than generic phrasing.

3. **ROUGE-L (Recall-Oriented Understudy for Gisting Evaluation)**

ROUGE-L focuses on recall by computing the longest common subsequence between the generated report and the reference. Unlike BLEU, which emphasizes precision, ROUGE-L is better suited to assess how much of the important content is preserved in the generated text, even if the phrasing is different. This metric is valuable in the clinical setting where capturing the full scope of findings is critical, even if the wording does not match exactly.

4. **CLIP-Score (Contrastive Language–Image Pretraining Score)**

CLIP-Score evaluates how well a generated report aligns semantically with its corresponding medical image by measuring similarity in a joint vision-language embedding space. Unlike traditional text-only metrics such as ROUGE or BLEU, CLIP-Score captures multimodal coherence by assessing whether the textual description is visually grounded. This is particularly important in radiology, where a generated report should not only be linguistically fluent but also reflect actual visual findings from the image. A higher CLIP-Score indicates that the report is more semantically aligned with the image, making it a valuable metric for assessing clinical relevance in vision-language generation tasks.

4 Results

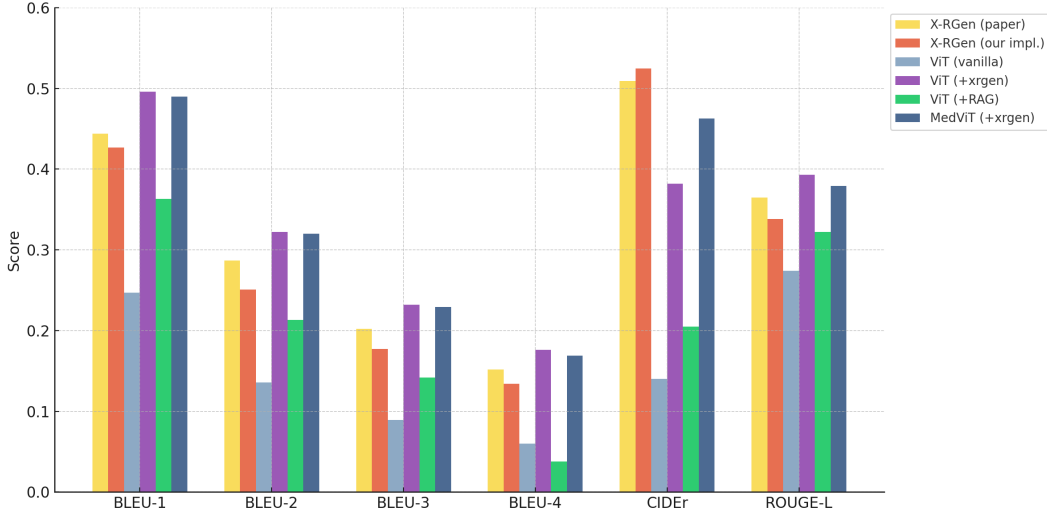


Figure 5: *Performance comparison of report generation models*

Table 1 and Figure 5 present a comparison of various models for radiology report generation using standard evaluation metrics. Our implementation of the X-RGen baseline performs comparably to the original paper, with slightly lower BLEU and ROUGE scores but a marginally higher CIDEr. The vanilla ViT model underperforms across all metrics, suggesting that Vision Transformers alone struggle to generate fluent and accurate reports without specialized guidance. However, integrating ViT into the X-RGen framework yields the best overall performance in BLEU and ROUGE metrics, highlighting the benefit of combining global visual features with structured report generation stages. Interestingly, while the ViT + RAG model boosts CIDEr compared to vanilla ViT, it shows a drop in BLEU-4, indicating possible factual improvements at the cost of syntactic precision. MedViT, when combined with the X-RGen framework, performs competitively, closely matching ViT + X-RGen, with slightly higher CIDEr and strong BLEU scores, suggesting its medical-specific pretraining provides valuable domain alignment. **These findings address RQ1**, demonstrating how different visual encoders and retrieval strategies impact report generation across multiple metrics.

Table 1: Performance comparison of different models on radiology report generation

Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	CIDEr	ROUGE-L
X-RGen (paper result)	0.444	0.287	0.202	0.152	0.509	0.365
X-RGen (our implementation)	0.427	0.251	0.177	0.134	0.525	0.338
ViT (vanilla)	0.247	0.136	0.089	0.060	0.140	0.274
ViT (+xrgen framework)	0.496	0.322	0.232	0.176	0.382	0.393
ViT (+RAG)	0.363	0.213	0.142	0.038	0.205	0.322
MedViT (+xrgen framework)	0.490	0.320	0.229	0.169	0.463	0.379

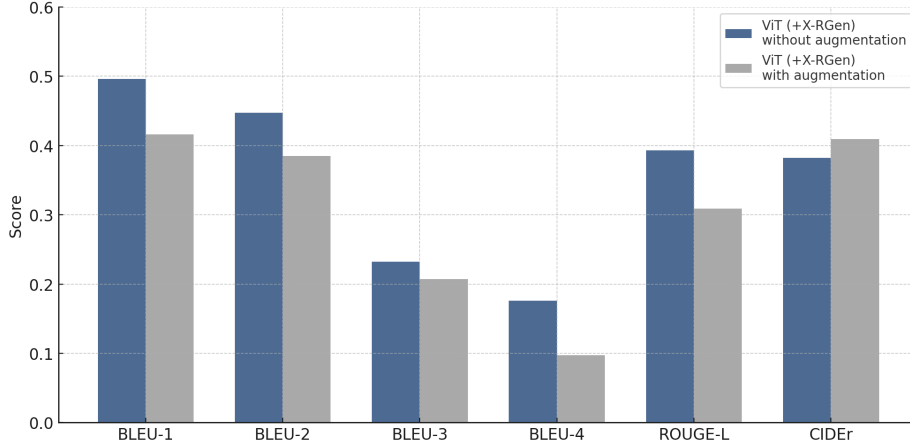


Figure 6: *Performance comparison of report generation models*

Table 2 and Figure 6 compare the performance of the ViT (+X-RGen) model with and without data augmentation. Interestingly, while data augmentation slightly improves the CIDEr score (0.409 vs. 0.382), indicating better factual relevance, it leads to drops in BLEU and ROUGE-L scores, especially at higher n-gram levels like BLEU-4 (0.097 vs. 0.176). This suggests that augmentation may have introduced variability that hindered fluency and precise n-gram overlap with reference reports, though it may have encouraged the model to capture diverse semantic content. Overall, these results highlight a trade-off: data augmentation may boost diversity and factual grounding but at the cost of syntactic precision. Fine-tuning augmentation strategies could help balance this trade-off more effectively. **This analysis addresses RQ2** by evaluating how RAG influences the balance between factual correctness and linguistic precision in report generation.

Table 2: Performance comparison of ViT (+X-RGen) with and without data augmentation

Model Setting	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-L	CIDEr
ViT (+X-RGen) <i>without augmentation</i>	0.496	0.447	0.232	0.176	0.393	0.382
ViT (+X-RGen) <i>with augmentation</i>	0.416	0.385	0.207	0.097	0.309	0.409

To answer RQ3, we also include CLIP-Score in our evaluation to assess the semantic alignment between the generated reports and the corresponding radiology images in Table 3. MedViT (+xrgen) achieves the highest CLIPScore (0.306), indicating the strongest semantic alignment between generated reports and images. X-RGen and ViT (+xrgen) perform comparably (0.295), suggesting limited benefit from simply adding the ViT into the X-RGen model. Interestingly, ViT (+RAG) underperforms slightly (0.281), implying that the retrieval module may not have effectively improved the relevance or may have introduced noise. Overall, MedViT’s performance shows promise for multimodal alignment in medical report generation.

Table 3: CLIP-Score across all Implementations

Evaluation	X-RGen	ViT (+xrgen)	MedViT (+xrgen)	ViT (+RAG)
CLIP-Score	0.296	0.295	0.306	0.281

5 Discussion

5.1 Limitations

We did face several challenges in the implementation. One major limitation was the computational overhead associated with large Transformer-based models such as GPT2 and BioClinicalBERT. Due to GPU memory constraints, we had to reduce the batch size and carefully manage resource allocation, especially when experimenting with larger decoders. Moreover, aligning the dimensionality between the ViT encoder output and the decoder’s input space required precise handling to avoid shape mismatches, especially when using only the CLS token for cross-attention. The evaluation posed subtle difficulties; while standard metrics like BLEU and CIDEr are useful, they do not fully capture clinical correctness or factual accuracy, highlighting the need for more domain-aware evaluation strategies in future work. We address these limitations through careful model tuning and dimensional alignment strategies. Moreover, we adapted our training setup to fit within memory constraints by optimizing batch sizes and leveraging modular design for flexible experimentation.

5.1.1 ViT+GPT Decoder

Vanilla ViT with the GPT-style decoder underperforms compared to X-RGen and MedViT variants. Given that vanilla ViT is a basic architecture pretrained on the ImageNet dataset and lacks inductive biases or features specific to medical imaging, this is to be expected. It does not encode spatial hierarchies or domain-specific visual patterns essential for radiology. Furthermore, unlike BioClinicalBERT in X-RGen, the GPT decoder lacks biomedical grounding. Together, this leads to shallow visual-textual alignment and less clinically coherent report generation.

5.1.2 MedViT

While MedViT achieved a notable improvement over vanilla ViT, when compared to the baseline that is the X-RGen implementation, it fell short. This suggests several important limitations that may have affected its performance in our setup.

First, MedViT is trained mostly on image classification tasks without any accompanying text or report data, even if it has been pretrained on a sizable corpus of medical images. In a text generation context, this restricts its capacity to generate features that are semantically consistent with language tokens. Unlike models pretrained with image-text pairs, MedViT lacks a mechanism for visual-textual grounding, which may explain the limited gains in CIDEr.

Although MedViT encoder and the decoder are trained end-to-end, the IU X-Ray dataset’s features could probably limit its performance [12]. The collection consists of only about 3,955 distinct radiology reports and about 7,470 photos, many of which are near duplicates (for example, frontal and lateral views of the same case). In order to properly fine-tune a big pretrained encoder like MedViT, this small and comparatively homogeneous dataset lacks the visual and linguistic diversity required. Moreover, the reports are prepared in a brief, semi-structured manner, which weakens the signal between textual targets and visual features. Due to class imbalance, rare conditions are underrepresented, making it more difficult for the model to learn accurate visual-to-language mappings.

We choose to use patch embeddings [8] as the input to the Transformer decoder in our architecture, removing the [CLS] token from the MedViT encoder. This is a common approach and illustrates how ViT has been strategically modified to meet the requirements of medical report generation [9]. Radiology reporting frequently depends on localized visual evidence, such as faint shadows, nodular structures, or effusions in certain lung zones, but the [CLS] token functions as a global summary in classification tasks. We make it possible for the model to more efficiently extract and translate spatially grounded findings by directing the decoder’s attention to patch-level tokens.

But there is a downside to this trade-off as well. By omitting the CLS token, we lose access to a holistic visual embedding that could benefit findings spanning multiple regions, such as cardiomegaly, hyperinflation, or diffuse interstitial patterns. The domain-specific global context of MedViT may be underutilized as a result, especially for summary-style lines that are frequently used in reports (e.g., "No acute cardiopulmonary abnormality").

Finally, according to our findings, even when pretrained on the ImageNet dataset images, the ViT+xrgen model gained more from the cross-modal design of the X-RGen framework than from domain-specific pre-training alone. This result is consistent with recent research, which indicates that joint training techniques and careful architectural alignment for multimodal fusion are frequently more important for downstream performance than pretraining on domain-specific data. Therefore, future work could be prioritizing developments in cross-modal interaction and training regimes to maximize the effectiveness of vision-language models in medical applications.

5.1.3 RAG

The RAG-enhanced model, which combines a ViT visual encoder with retrieval-augmented generation, outperforms the vanilla ViT baseline across all metrics. This demonstrates the potential of retrieval in enriching the input with semantically related context. However, its performance remains lower than the ViT model integrated within the XRGen framework. Several factors could explain this partial improvement.

First, while the retrieved reports provide additional context, they may also introduce semantic noise. [13] Retrieved reports may not always contain directly relevant or complementary information. In radiology, subtle findings and patient-specific language make retrieved examples less generalizable. If the model is not equipped with a mechanism to weigh the relevance of retrieved content (e.g., gated attention or retrieval confidence scores), it may treat all retrieved text equally, even when it is irrelevant or misleading. This can confuse the decoder and degrade the generation quality.

Second, prepending the retrieved text consumes part of the decoder’s fixed input length budget [15] [14]. This forces truncation of the ground-truth or the retrieved text, potentially discarding important training signals. Since the decoder receives a concatenated sequence of retrieved context and target report, it may struggle to differentiate between what is context and what is expected to be generated. Without fine-grained control or a separate encoder for the retrieved memory, the model treats both as a single stream of tokens, which can dilute learning.

Third, the retrieval process is frozen during training. The ViT embeddings used for retrieval are not updated jointly with the decoder, meaning the retrieval quality does not adapt or improve as training progresses [13]. This is especially limiting in a dataset like IU X-Ray, where visual variability is low and retrieval can often return semantically redundant or overly generic reports [2]. Moreover, RAG is only applied during training in this setup and during testing, the model relies solely on the input image and does not have access to the retrieval context. This train-test mismatch could lead to inconsistent behavior during inference.

Lastly, the XRGen framework is specifically designed to optimize image-to-text alignment through contrastive and reconstruction objectives, using joint vision-language representation learning. In contrast, RAG does not adaptively align retrieved content with the image during training, making it less robust in structured medical domains. While RAG improves upon plain ViT by offering broader context, it lacks the targeted alignment and architectural tuning that make XRGen more effective in radiology report generation.

5.2 Future Work

While the current work focuses on using Vision Transformers (ViTs) in place of convolutional backbones and investigating their potential for generating radiological reports, there are still a number of interesting avenues for further research. A key area could be to make greater use of the [CLS] token to integrate

global context with localized findings, either by fusing it with patch-level features or by using it to initialize decoder states [10][11]. Furthermore, adding cross-modal pretraining goals, such as contrastive learning with image-report pairs, may improve alignment and visual-textual grounding. Researching multiview fusion techniques to better combine frontal and lateral X-rays could help enhance performance in spatial reasoning scenarios. Also, leveraging structured clinical priors or scaling to bigger datasets like MIMIC-CXR [15] may help domain-pretrained encoders like MedViT reach additional potential. We could also integrate explainability methods for the model like attention map visualization or natural language rationale that highlight which image regions contributed to specific report statements helping trust and clinical adoption.

As part of our future work, we plan to expand the evaluation beyond standard NLP metrics by incorporating clinical NLP tools such as MedSpaCy to assess the structural completeness and clinical relevance of the generated radiology reports. While current metrics like BLEU, CIDEr, and ROUGE-L provide a general sense of linguistic fluency and content overlap, they fall short in capturing whether the generated reports contain the essential clinical components required for medical decision-making. By leveraging MedSpaCy, we aim to extract structured entities such as findings, anatomical locations, negations, and diagnoses, and compare them against ground-truth reports.

5.3 Conclusion

In this work, we explored the integration of Vision Transformers (ViTs) into radiology report generation pipelines, replacing traditional convolutional backbones within the X-RGen framework. Using the IU X-Ray dataset, we systematically evaluated vanilla ViT, MedViT, and a ViT integrated with a retrieval augmented generation (RAG) setup. Our experiments show that while ViT encoders bring modularity and flexibility, they require careful adaptation to the medical domain. Vanilla ViT underperforms due to its lack of domain bias and simplistic architecture. MedViT, on the other hand, introduces valuable inductive priors but is limited by dataset scale and the absence of cross-modal pretraining objectives.

We also use a Retrieval-Augmented Generation (RAG) approach to tackle the problem of factual grounding and semantic richness in generated reports. We increase the informativeness and coherence of the outputs produced by using FAISS to get semantically related training reports based on image embeddings and then feeding them into the decoder as auxiliary context. A possible avenue for improving radiology report quality through the use of memory-augmented transformer models is highlighted by this modification of RAG for multimodal, free-form clinical production. To further enhance generalization, we experimented with data augmentation strategies, which showed a trade-off between improved factual content and reduced syntactic precision.

To complement standard metrics like BLEU and ROUGE, we incorporated CLIPScore to better assess semantic alignment between images and generated reports, addressing known limitations of traditional metrics in capturing clinical and factual correctness.

For **RQ1**, our results show that replacing CNN encoder with ViT or MedViT encoders in the X-RGen framework improves report generation quality, with MedViT achieving strong domain-aligned performance across BLEU, ROUGE, and CIDEr metrics. For **RQ2**, while integrating retrieval (RAG) improves factual accuracy, as indicated by CIDEr, it can reduce syntactic fluency, revealing a trade-off between semantic richness and textual precision. For **RQ3**, CLIP-Score evaluation demonstrates that MedViT (+X-RGen) achieves the highest semantic alignment with radiology images, confirming the benefit of domain-specific vision-language modeling for multimodal consistency.

Overall, our work provides a principled evaluation of ViT-based encoders in medical vision-language tasks, identifies architectural trade-offs around global context and fusion, and proposes a retrieval-enhanced pipeline to enrich semantic alignment. These contributions lay the foundation for future research in scalable, domain-aware, and fact-grounded generative medical AI systems.

6 Team Membership and Attestation

Team members Madhumitha V, Sriharshini Dusi, Yueqiao Chen participated sufficiently.

Code & Data Availability Statement

Code: Github Repository

Data: UI-XRay Dataset [2]

References

- [1] Chen, Q., Xie, Y., Wu, B., Chen, X., Ang, J., To, M.-S., Chang, X., & Wu, Q. (2024). Act Like a Radiologist: Radiology Report Generation Across Anatomical Regions. In *Computer Vision – ACCV 2024: 17th Asian Conference on Computer Vision*, December, 2024, Proceedings, Part VI (pp. 36–52). https://doi.org/10.1007/978-981-96-0960-4_3
- [2] Demner-Fushman, D., Kohli, M. D., Rosenman, M. B., Shooshan, S. E., Rodriguez, L., Antani, S., Thoma, G. R., & McDonald, C. J. (2016). Preparing a collection of radiology examinations for distribution and retrieval. *Journal of the American Medical Informatics Association*, 23(2), 304–310. <https://drive.google.com/file/d/1c0BXEuDy8Cmm2jfn0YYGkQxFZd2ZIoLg/view?pli=1>
- [3] Li, Y., Kumar, A., Singh, R., & Zhang, Y. (2025). A Survey of Deep Learning-based Radiology Report Generation Using Multimodal Data. *arXiv preprint*, arXiv:2501.01234. <https://arxiv.org/abs/2405.12833>
- [4] Zhou, M., Lee, J., Wang, R., & Patel, V. (2024). Multi-modal Transformer Architecture for Medical Image Analysis and Automated Report Generation. *IEEE Transactions on Medical Imaging (under review)*. [@inproceedingsranjit2023retrieval, title=Retrieval augmented chest x-ray report generation using openai gpt models, author=Ranjit, Mercy and Ganapathy, Gopinath and Manuel, Ranjit and Ganu, Tanuja, booktitle=Machine Learning for Healthcare Conference, pages=650–666, year=2023, organization=PMLR](https://www.researchgate.net/publication/383266445_Multi-modal_transformer_architecture_for_medical_image_analysis_and_automated_report_generation)
- [5] Wu, Jinge and Wu, Zhaolong and Li, Ruizhe and Hasan, Abul and Kim, Yunsoo and Cheung, Jason P.Y. and Zhang, Teng and Wu, Honghan(2024). Integrating Knowledge Retrieval and Large Language Models for Clinical Report Error Correction. *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*.<https://arxiv.org/pdf/2406.15045>
- [6] Chen, M., Zhang, Z., Huang, Z., Chen, Y., Tan, Y., & Zhang, Y. (2021). Generating Radiology Reports via Memory-Driven Transformer. *IEEE Transactions on Medical Imaging*, 40(4), 1188–1199.<https://arxiv.org/abs/2010.16056>
- [7] Tang, A., Phang, J., Le, E., Banerjee, I., & Lungren, M. (2024). Enhancing Large Language Models with Retrieval-Augmented Generation: A Radiology-Specific Approach. *Radiology: Artificial Intelligence*, 6(3), e240313. <https://pubs.rsna.org/doi/10.1148/ryai.240313>
- [8] Wang, X., Figueredo, G., Li, R., Zhang, W. E., Chen, W., & Chen, X. (2025). A survey of deep-learning-based radiology report generation using multimodal inputs. *Expert Systems with Applications*, 289, 128328. <https://www.sciencedirect.com/science/article/pii/S1361841525001744>

- [9] Sun, D., Mu, C., Fan, X., Tang, J., Li, Z., Zhu, J., & Ding, Z. (2025). Hybrid graph-based radiology report generation. *Expert Systems with Applications*, 289, 128174. <https://www.sciencedirect.com/science/article/abs/pii/S0957417425019475>
- [10] Hou, C., Zhou, Y., Feng, J., & others. (2021). CrossViT: Cross-Attention Multi-Scale Vision Transformer for Image Classification. *arXiv preprint arXiv:2103.14899*. <https://arxiv.org/pdf/2103.14899>
- [11] Li, Y., Wang, H., Liu, X., & others. (2025). Fusion feature-based hybrid methods for diagnosing oral squamous cell carcinoma using CrossViT. *Frontiers in Oncology*, 15, 1551876. <https://www.frontiersin.org/journals/oncology/articles/10.3389/fonc.2025.1551876/full>
- [12] Harzig, P., Chen, Y.-Y., Chen, F., & Lienhart, R. (2020). Addressing Data Bias Problems for Chest X-ray Image Report Generation. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2020. University of Augsburg, Germany and FX Palo Alto Laboratory, USA. <https://arxiv.org/pdf/1908.02123>
- [13] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Kiela, D., Moens, M.-F., & Riedel, S. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *Advances in Neural Information Processing Systems*, 33, 9459–9474. <https://dl.acm.org/doi/abs/10.5555/3495724.3496517>
- [14] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention Is All You Need. *Advances in Neural Information Processing Systems*, 30. <https://arxiv.org/abs/1706.03762>
- [15] Johnson, A. E. W., Pollard, T. J., Berkowitz, S. J., Greenbaum, N. R., Lungren, M. P., Deng, C. Y., Mark, R. G., & Horng, S. (2019). MIMIC-CXR: A Large Publicly Available Database of Labeled Chest Radiographs. *arXiv preprint arXiv:1901.07042*. <https://arxiv.org/pdf/1901.07042>
- [16] Ranjit, M., Ganapathy, G., Manuel, R., & Ganu, T. (2023). Retrieval Augmented Chest X-ray Report Generation Using OpenAI GPT Models. *Machine Learning for Healthcare Conference*, 650–666. <https://proceedings.mlr.press/v211/ranjit23a.html>